Type MS · Sign and magnitude numbers

Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · Sign and magnitude numbers

Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
• Equivalent expressions are accepted
• Calculations not explicit but expressed gain the mark
• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td colspan="2"></td><td>Total for question 2</td><td>15</td></tr>
</table>

Type MS · Sign and magnitude numbers

Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sign and magnitude numbers

Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part e.i in this mark scheme extract

3(a)(i)	• LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• WAN / Wide area network (1)		1

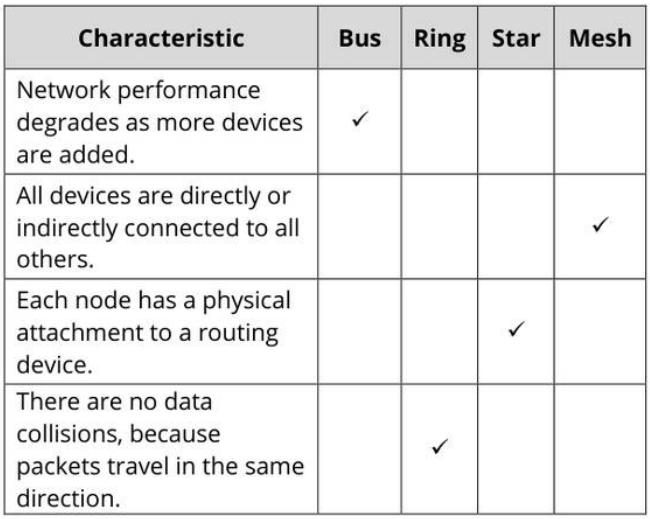
Question Number	Answer	Additional Guidance	Mark
3(b)	A: Megabits per second **The only correct answer is A** **B** is not correct because megapixels are associated with images rather than network data speeds **C** is not correct because mebicycles are not associated with network data speeds **D** is not correct because mebihertz are not associated with network data speeds		1

Type MS · Sign and magnitude numbers

Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

Question Number	Answer	Additional Guidance	Mark
3(d)			4

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	✓			
All devices are directly or indirectly connected to all others.				✓
Each node has a physical attachment to a routing device.			✓	
There are no data collisions, because packets travel in the same direction.		✓		

Type MS · Sign and magnitude numbers

Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Sign and magnitude numbers

Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Sign and magnitude numbers

Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

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Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

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Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sign and magnitude numbers

Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d.ii in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

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Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · Sign and magnitude numbers

Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · Sign and magnitude numbers

Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · Sign and magnitude numbers

Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Sign and magnitude numbers

Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · Sign and magnitude numbers

Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

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Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark